

e Knowledge tasks, intervention pathways and action-stage reach

Intervention residual fingerprint

NSF-NSFC conditional share difference (percentage points)

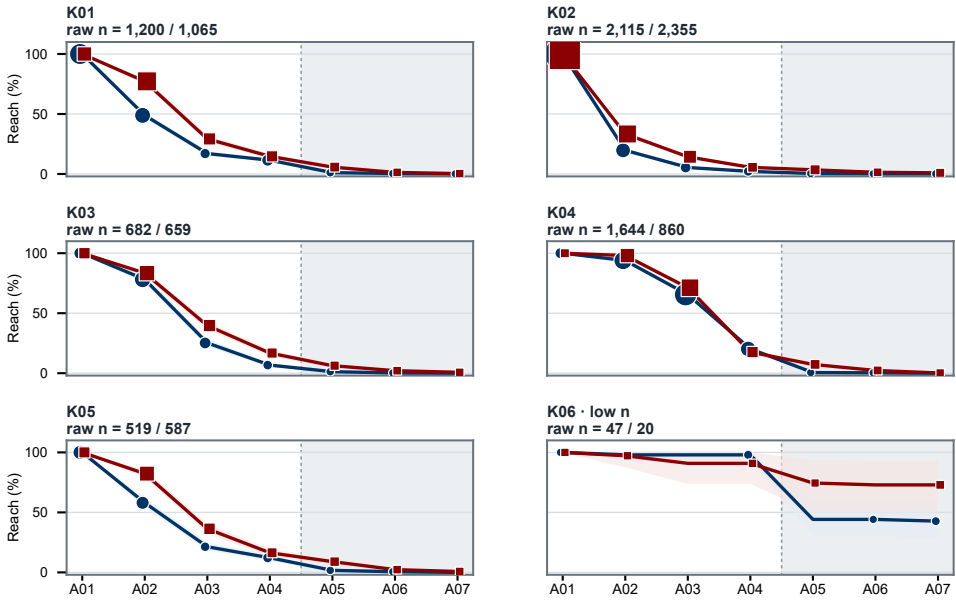


● NSF higher ● NSFC higher ○ 95% CI excludes 0

Absolute conditional-share difference
○ 10 pp ○ 20 pp ○ 30 pp

Action-stage reach by knowledge task

curve = balanced P(stage ≥ A); bubble area = raw exact-stage project n



A05–A07 shaded: decision integration, diffusion/scaling and post-evaluation

— NSF — NSFC ■ project-bootstrap 95% CI

Exact-stage projects
● n=10 ● n=100 ● n=1,000

Knowledge tasks

K01 State measurement and problem diagnosis
K03 Scenario simulation and risk prediction
K05 Planning decision support and policy design

K02 Mechanism analysis and causal identification
K04 Intervention design and performance optimization
K06 Implementation monitoring and impact evaluation

Intervention types

L01 Engineering and infrastructure intervention
L04 Spatial planning intervention
L07 Legal and institutional intervention

L02 Digital technology intervention
L05 Public service and operational intervention
L08 Fiscal, financial and market intervention

L03 Ecological restoration intervention
L06 Behavioural and demand-management intervention
L09 Participation, collaboration and capacity-building intervention